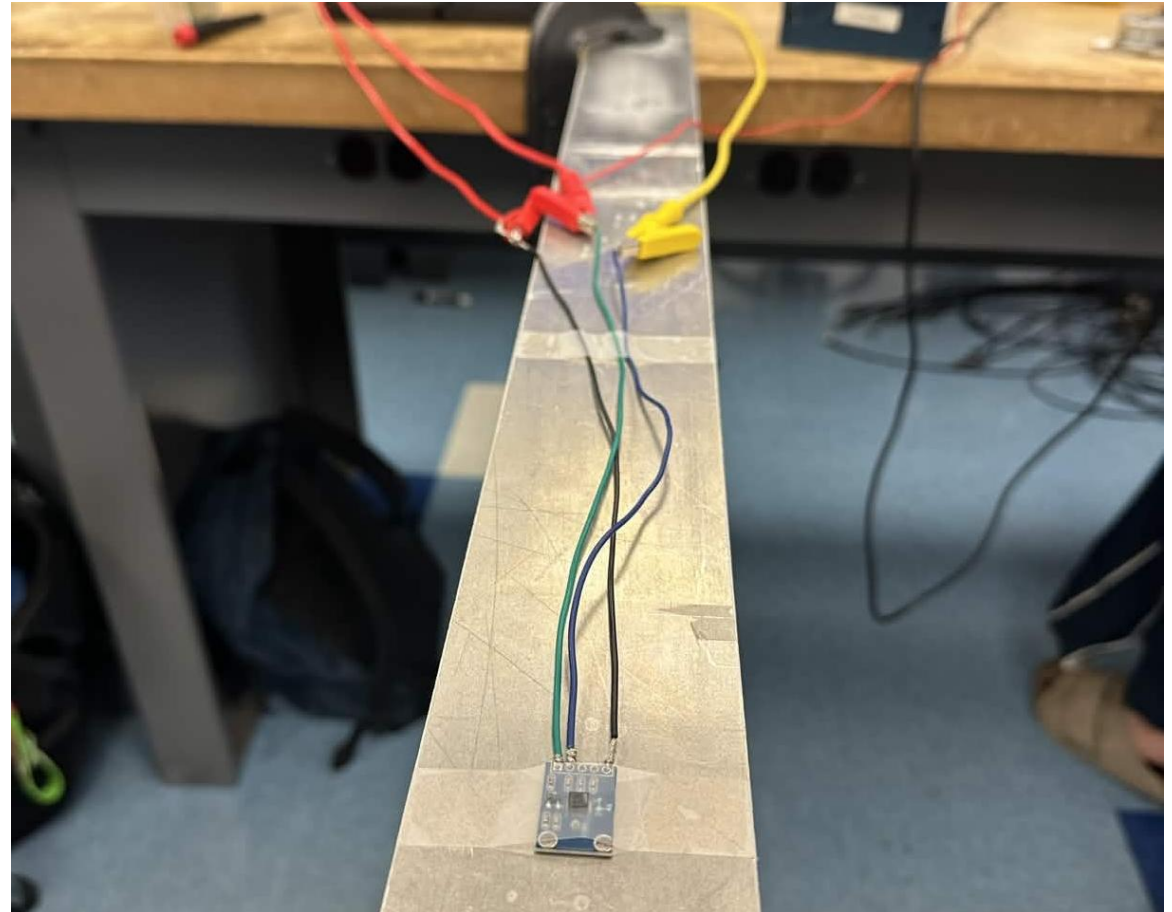


Cantilever Beam Vibration Project

Using an accelerometer, NI USB-6341 DAQ, and LabVIEW to compare beam response under added mass

Conditions tested: No mass, 1 lb, 3 lb, and 5 lb



Presentation Roadmap

1. Objective and theory

What a cantilever beam is, why vibration changes with mass, and what we expected before testing.

2. Equipment and setup

Accelerometer, DAQ, LabVIEW, beam mounting, and the sensor wiring used to collect data.

3. Procedure

How each trial was recorded for the no-mass, 1 lb, 3 lb, and 5 lb conditions.

4. Raw data

Voltage-versus-time responses for each loading condition and what the signals looked like.

5. Calculations

Estimating period from recorded peaks and computing frequency using $f = 1/T$.

6. Results, limits, conclusion

How added mass changed response, limitations in the experiment, and the final takeaway.

Objective and Expected Behavior

Objective

Measure the cantilever beam response after excitation and compare how the period and frequency change when 1 lb, 3 lb, and 5 lb are added.

Expected trend

A larger attached mass increases the effective inertia of the beam system, so the oscillation period should increase and the dominant frequency should decrease.

Key ideas used in the analysis

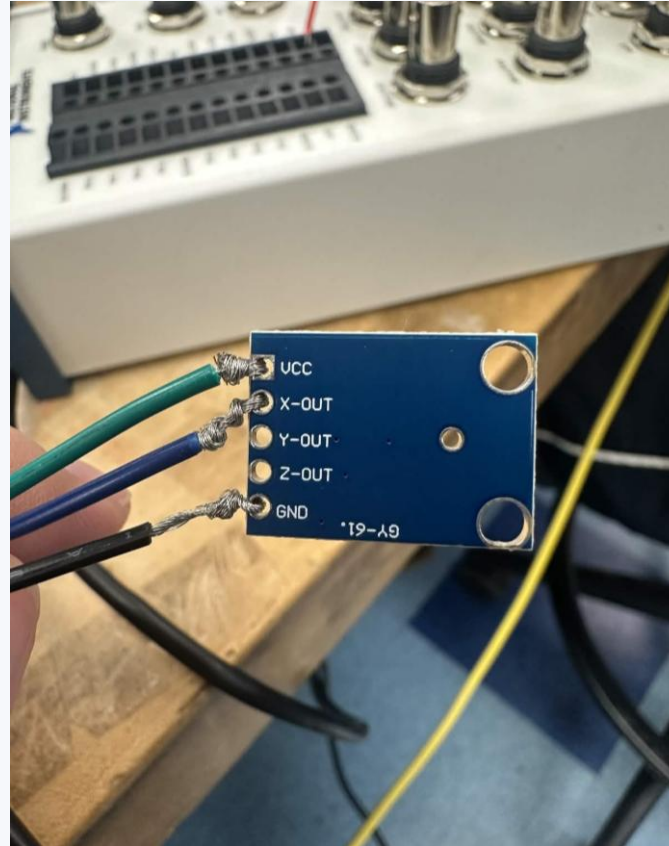
- The beam was excited and then allowed to vibrate freely.
- The accelerometer output was recorded as voltage versus time.
- The time between consecutive peaks was used to estimate the period T .
- Frequency was then calculated from the measured period using $f = 1/T$.
- The results were compared across no mass, 1 lb, 3 lb, and 5 lb.

$$\text{Frequency equation: } f = 1/T$$

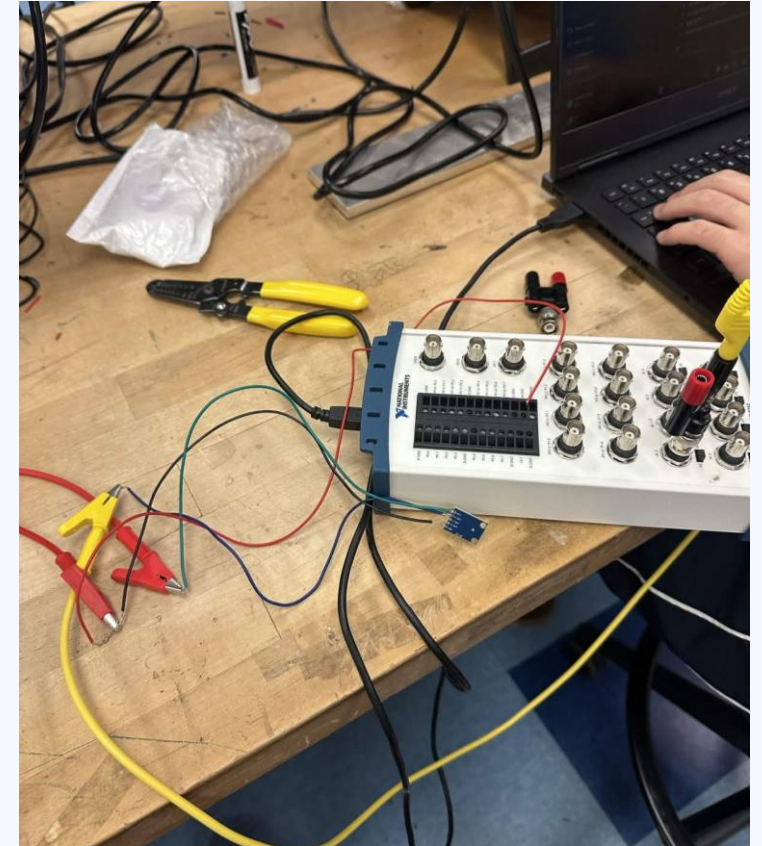
Equipment Used

Main equipment

- Metal cantilever beam
- Accelerometer mounted near the free end
- National Instruments USB-6341 DAQ
- Computer running LabVIEW
- Connecting wires and clips
- Added masses: 1 lb, 3 lb, and 5 lb



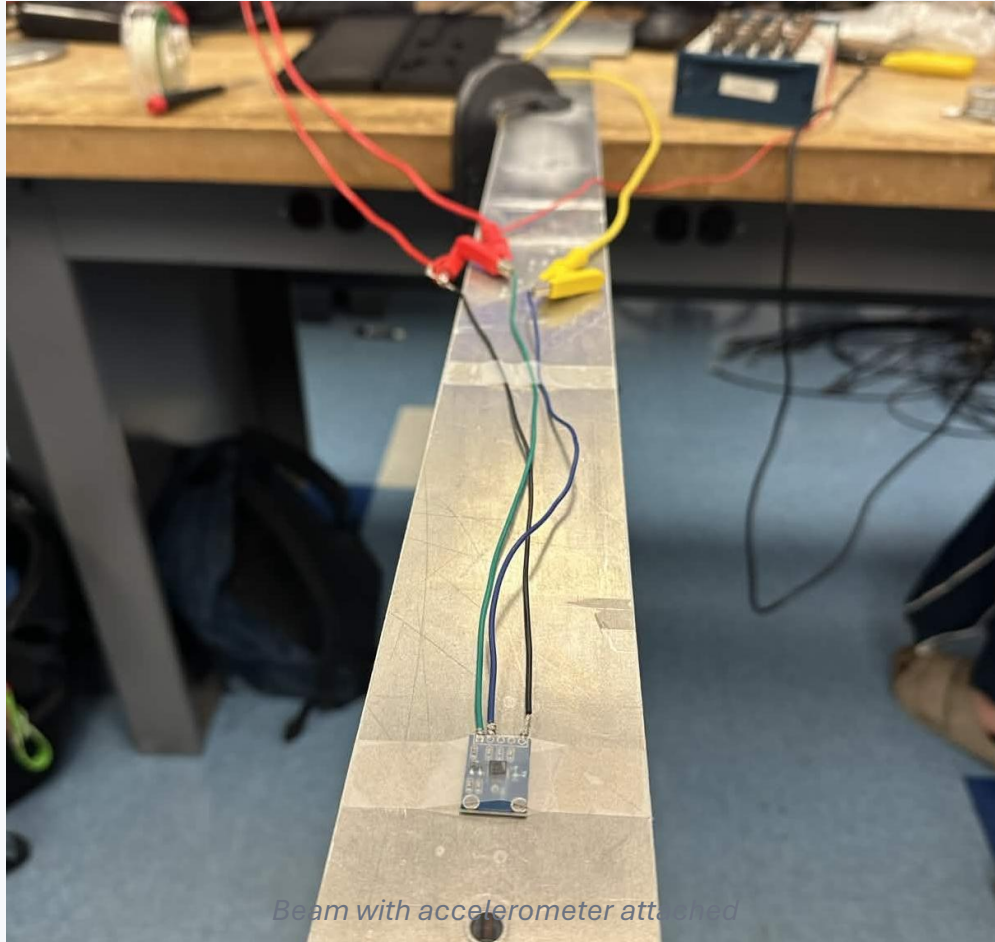
Accelerometer wiring



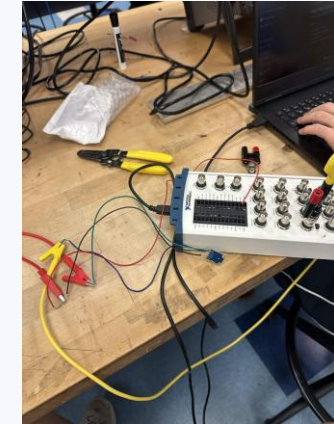
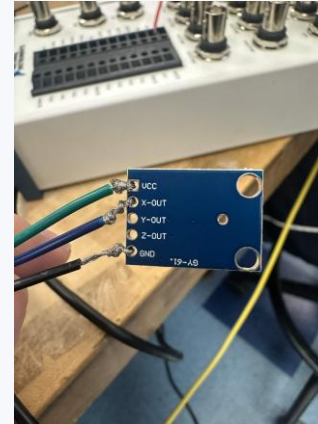
NI USB-6341 DAQ setup

Experimental Setup

Photos of the beam, sensor wiring, and DAQ connection used during testing.



Beam with accelerometer attached



How the system worked

The accelerometer converted the beam motion into a voltage signal. The DAQ sent that signal into LabVIEW, where each trial was recorded as a time-based measurement file.

Procedure and Data Collection

Each condition used the same process so the comparisons would be fair.

Step 1

Mount the accelerometer on the beam and connect VCC, signal output, and ground to the DAQ.

Step 2

Confirm that LabVIEW is reading a stable signal before exciting the beam.

Step 3

Excite the beam and record the response as voltage versus time.

Step 4

Repeat for no mass, 1 lb, 3 lb, and 5 lb. Save each trial as an LVM file.

Step 5

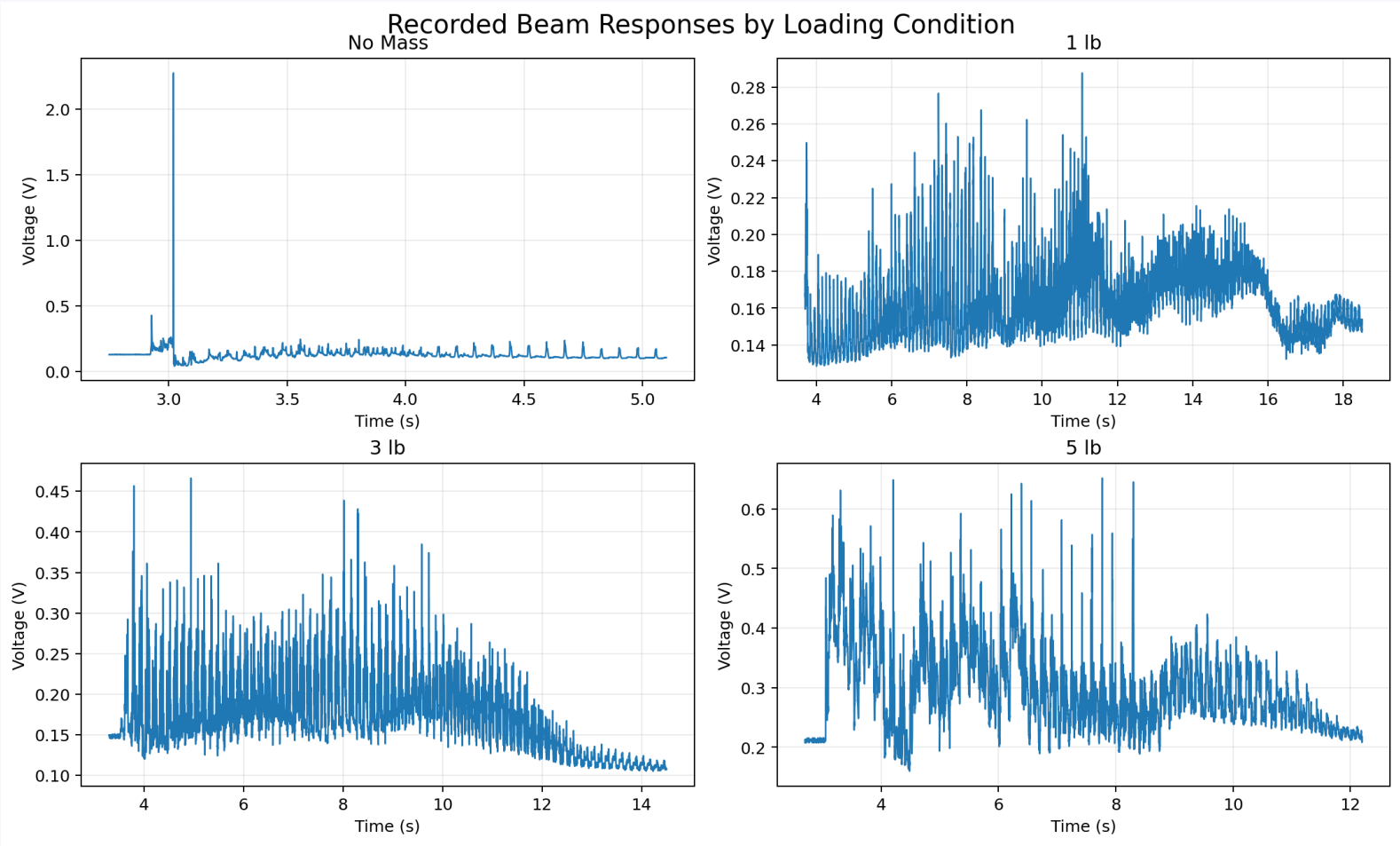
Use the recorded peaks to estimate period and then calculate frequency.

Sampling details

The LabVIEW files used a sampling interval of 0.001 s, which corresponds to a 1000 Hz sampling rate. That gave enough time resolution to identify the main oscillation peaks and compare the four loading conditions.

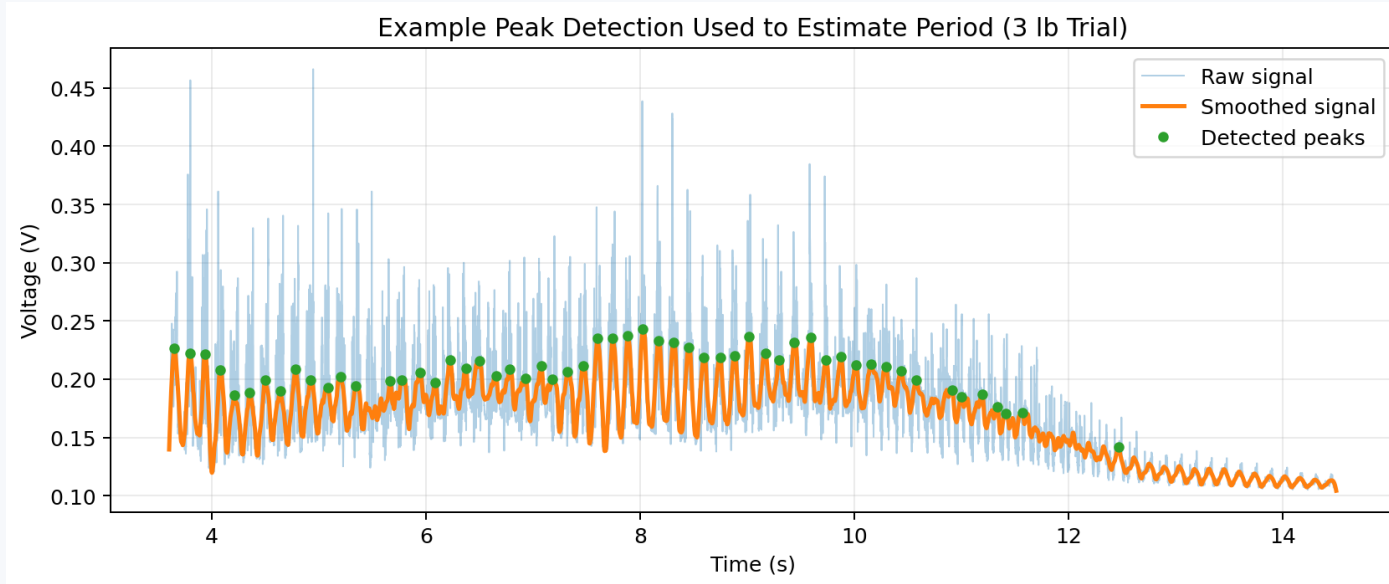
Raw Recorded Responses

Each plot shows the voltage response after the beam was excited.



Estimating Period and Frequency from Peaks

We identified peaks in the post-excitation region and measured the spacing between them.



Calculation method

1. Smooth the signal enough to identify the main peaks.
2. Measure the time between consecutive peaks.
3. Average those time differences to estimate T .
4. Compute frequency from $f = 1/T$.

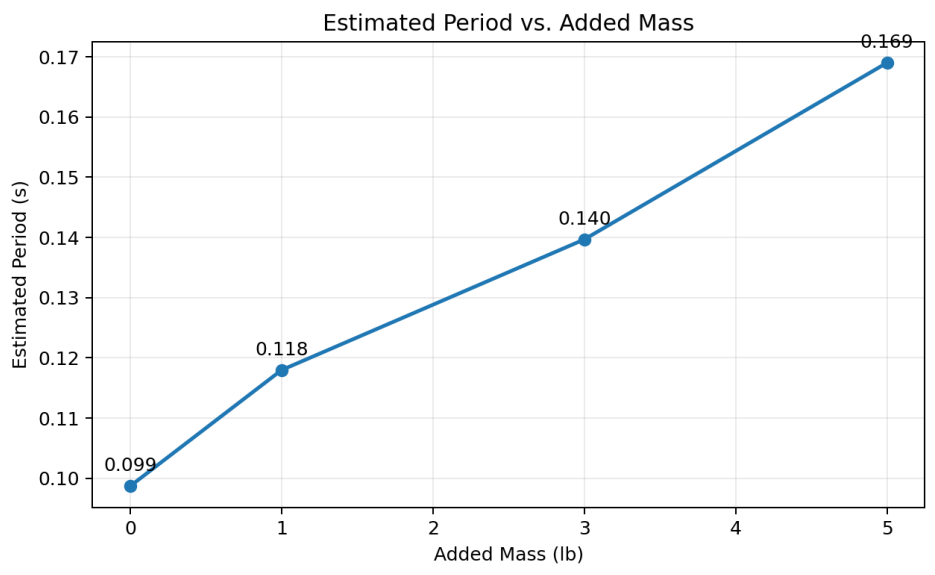
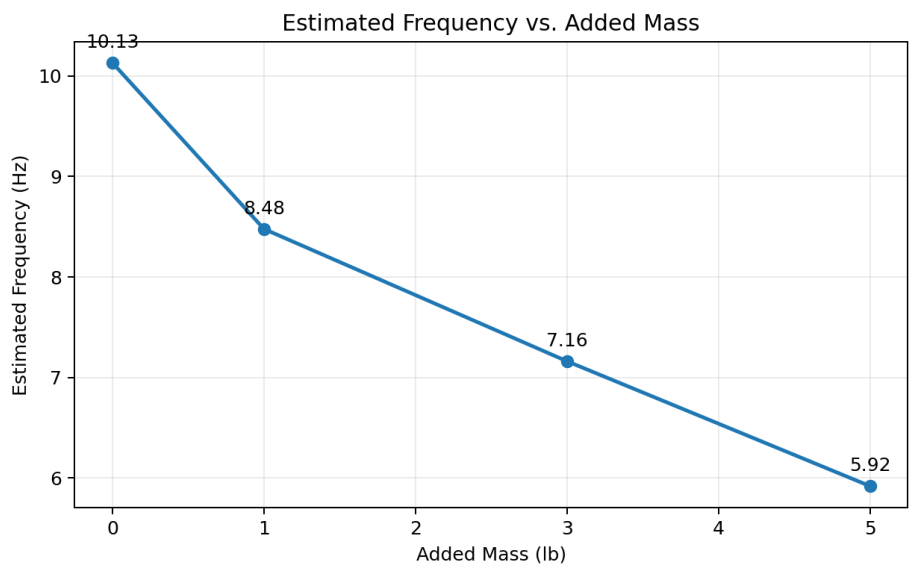
Example

For the 3 lb case, the average spacing between successive peaks was about 0.140 s. Using $f = 1/T$ gives an estimated frequency of about 7.16 Hz.

Processed Results

The period increased and the frequency decreased as more mass was added.

Condition	Mass (lb)	Estimated Period (s)	Estimated Frequency (Hz)	Approx. Peak-to-Peak Voltage
No Mass	0	0.099	10.13	0.204
1 lb	1	0.118	8.48	0.159
3 lb	3	0.14	7.16	0.361
5 lb	5	0.169	5.92	0.492



What the Results Mean

The measured trend matched the expected physical behavior of the beam.

Frequency trend

Estimated frequency dropped from about 10.13 Hz with no mass to about 5.92 Hz at 5 lb. That means the beam oscillated more slowly as load increased.

Signal amplitude

The heavier cases also showed larger peak-to-peak response in our recorded signals, especially the 3 lb and 5 lb trials.

Takeaway statement

Our measurements support the expected conclusion for a cantilever beam: adding mass slows the vibration response. The exact values are specific to our setup, but the overall trend is consistent with beam vibration theory.

Period trend

Estimated period increased from about 0.099 s with no mass to about 0.169 s at 5 lb, which means each cycle took longer with heavier loading.

Overall answer

Added mass clearly changed the beam response. Based on the processed peaks, more mass caused a lower dominant frequency and a longer oscillation period.

Limitations and Sources of Error

These factors help explain why the signals were somewhat noisy and why repeated trials would improve confidence.

- Some trials contained electrical noise and abrupt spikes, which made the signals less clean than an ideal damped sinusoid.
- The force used to excite the beam was not perfectly identical for every trial, so the starting amplitude varied.
- Only one main usable trial was used for each mass condition, which limits repeatability.
- The exact location and attachment of the masses may have introduced small differences between trials.
- The frequency values are estimates based on recorded peaks, so they should be interpreted as approximate experimental results rather than exact theoretical values.

Future improvement: collect multiple trials for each mass and average the measured periods and frequencies.

Conclusion

Final answer to the project question

What we accomplished

We successfully used the accelerometer, NI USB-6341 DAQ, and LabVIEW to measure beam vibration under four loading conditions.

Calculated results

From the recorded peaks, we estimated both period and frequency for the no-mass, 1 lb, 3 lb, and 5 lb cases.

Main conclusion

As mass increased, the estimated period increased and the estimated frequency decreased. This means the beam vibrated more slowly when additional mass was attached.

Questions?